PROPHEUS

How a Global Snack Manufacturer Unlocked $48M in Revenue with AI-Powered Product Recommendations

1 Scenario

For one of the world's leading snack manufacturers, product sales were being driven by a familiar pattern: a handful of best-selling SKUs accounted for the majority of orders. Some products were flying off the shelves; others, equally promising, were never even recommended to retailers - let alone stocked.

The company had built a robust distribution network across thousands of retail outlets, supported by a large on-ground sales team. But despite the breadth of their footprint, **sales reps were defaulting to the same handful of "safe" SKUs** across all stores. Their decisions were guided by limited historical sales data and anecdotal, informal knowledge - insights shared between reps or drawn from personal experience. This resulted in one-size-fits-all selling, missed cross-sell opportunities, and a lack of SKU diversity at the shelf.

2 The Cold Start Problem: When History Isn't Enough

The challenge wasn't just limited insight—it was a structural data gap. The company had no historical data connecting which SKUs were recommended to a retailer and whether those recommendations led to purchases. In technical terms, this created a cold start problem.

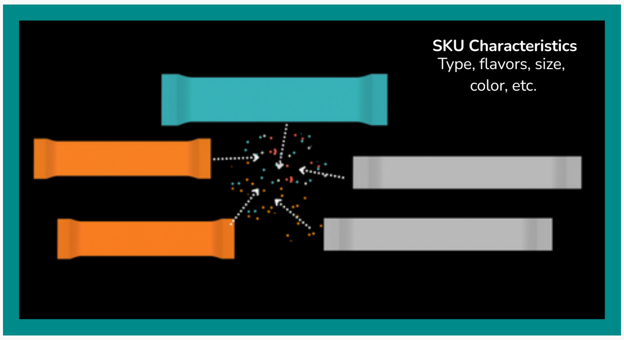
The goal was to recommend new or under-utilized SKUs to the right retailers, even if there was no prior sales relationship. But without a feedback loop of "recommended vs. bought," traditional recommendation systems would struggle.

This was the starting point when the company partnered with us to deploy **Propheus' Product Recommendation Agent.**

3 The Approach: Learning Without Recommendation History

To address the cold start challenge, we couldn't rely on collaborative filtering in the traditional sense. So we built a model that could learn from features, not just past interactions.

We began by encoding every SKU with a rich vector of attributes—category, pack size, brand, historical performance, and even visual elements like the wrapper color, text placement, and presence of brand ambassadors. These embeddings helped the model understand product similarity and appeal, even for SKUs with limited sales exposure.



At the same time, we built equally detailed embeddings for retailers. Each outlet was represented by a unique signature that captured not just purchase behavior and store type, but also real-world context: foot traffic patterns, nearby businesses, regional demographics, and local events - sourced from our proprietary Digital Atlas.

The model's task was to learn a function that could estimate the likelihood of a retailer purchasing a particular SKU based on the interaction between product and retailer features.

For instance, if Retailer A had previously purchased a 14g strawberry wafer and the system detected that a 16g vanilla wafer shared 95% of the same characteristics (from price point to customer profile), it could recommend the new SKU with confidence - even without any sales history between that retailer and the product.

The Model: Built on 370+ Features and Real-World Signals

The final model learned from over 370 features across five key categories:

Retailer Behavior (purchase history, basket size, product variety)	**SKU Attributes** (category, brand, packaging)	**Sales Trends** (Regional and outlet-level performance)	**Market & Geographic Signals** Store reach, surrounding businesses)	**Mobility Data & Local Context** (from the Digital Atlas)

Instead of pushing top-performing SKUs everywhere, the engine could **intelligently match the right product to the right outlet**, accounting for both product potential and local relevance.

This wasn't just a backend intelligence layer. Sales reps received smart, ranked SKU lists for each store—complete with contextual explanations (e.g., "high local footfall," or "similar outlets performing well") and one-line sales pitches customized to the retailer's profile.

Evaluation: Ranking Instead of Backtesting

Because there was no historical "recommended vs. bought" data, we couldn't test accuracy the traditional way. So we evaluated the model based on **how well it ranked retailer preferences**. For instance, if Retailer A had historically bought:

- 8 units of S2
- 6 units of S1
- 5 units of S3

And the model ranked S2 highest, followed by S1 and S3, we could conclude it had learned the underlying demand pattern.

4 The Results: $48M Revenue Uplift & More Balanced Sales

Once deployed, the impact was immediate and measurable. For a company with ~$600M in annual revenue, Propheus delivered a **net yearly impact of $48M** in revenue.

- **31% of recommended SKUs were purchased**, demonstrating a high hit rate on recommendations
- Retailers increased their **average basket size by 8%**
- This led to a **12% uplift in store-level revenue**

Perhaps most importantly, the model helped **rebalance SKU distribution**. While historical sales were concentrated on a few SKUs, the AI-driven recommendations introduced greater diversity—bringing lesser-known but relevant products into the picture.

A New Way to Sell

With Propheus, the sales reps became **data-augmented advisors**, equipped with real-time intelligence and context-rich recommendations. The model didn't just react to past behavior. It anticipated opportunity. It saw what humans couldn't. And it delivered results at scale. This case highlights how AI, when designed around real-world complexity, can go beyond automation - it can **amplify judgment, sharpen execution, and unlock growth** that legacy tools leave on the table.

If you'd like to see this solution in action for your business, reach out to **michelle@propheus.com** for more info!